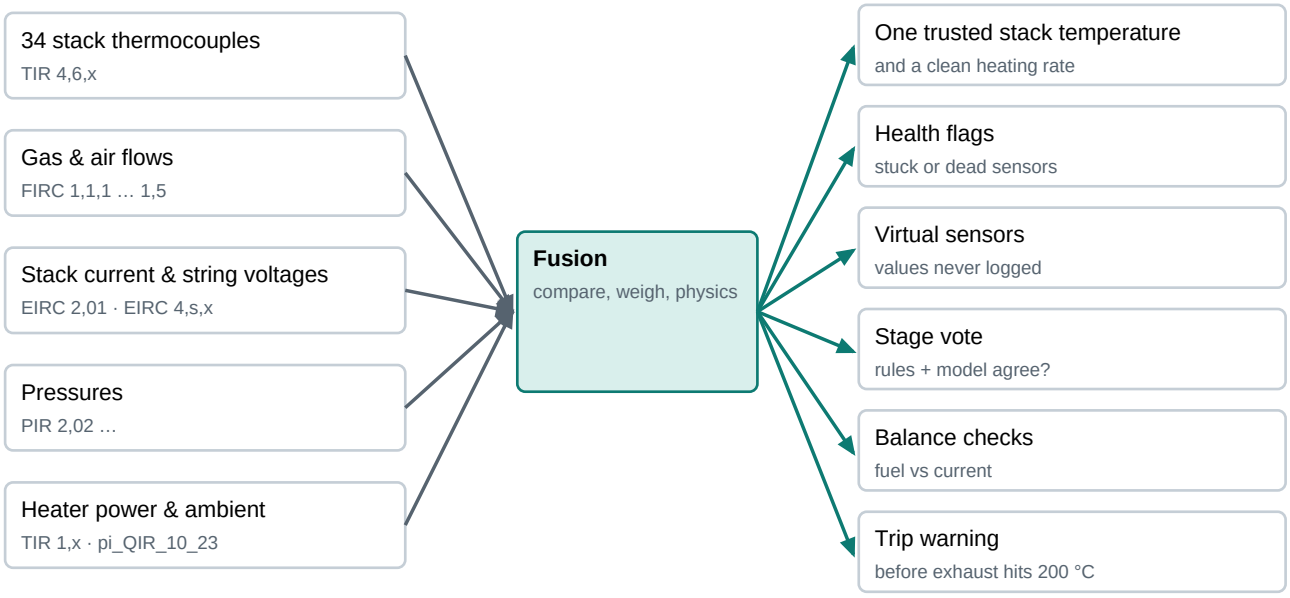
SOFC project · 16 Oct 2026 · companion to the web pages

How can sensor fusion help this fuel-cell project?

Short answer: sensor fusion means letting several sensors check and fill in for each other, so the plant data you train on and fake from is more trustworthy than any single sensor. In this project it gives six concrete wins, and three of them can be built from data we already have.

Why it matters: our 39 days of logs have a stuck thermocouple, 40 dead package-voltage readings, one sensor never logged, logger dropout minutes, and very little part-load data. Every later step inherits those flaws. Fusion is the layer that catches and repairs them.

Plain gloss. Think of a jury: one witness can be wrong, but twelve who mostly agree are hard to fool, and the one who disagrees stands out. Sensor fusion is that jury for plant sensors, plus a referee who knows physics.



The plant already measures the same things in several ways; fusion combines them into six outputs.

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IF YOU REMEMBER ONE THING

Sensor fusion turns many imperfect sensors into one set of numbers you can trust, and it is the natural quality layer under the fake-data generator.

Three ways to combine sensors

Fusion can happen on the raw readings, on summary features, or on final decisions. This project can use all three.

Level 1 · Raw readings
Combine readings of the same thing.

34 stack thermocouples → one stack temperature (drop the stuck one, take a robust middle).

Level 2 · Features
Combine different signals into a picture.

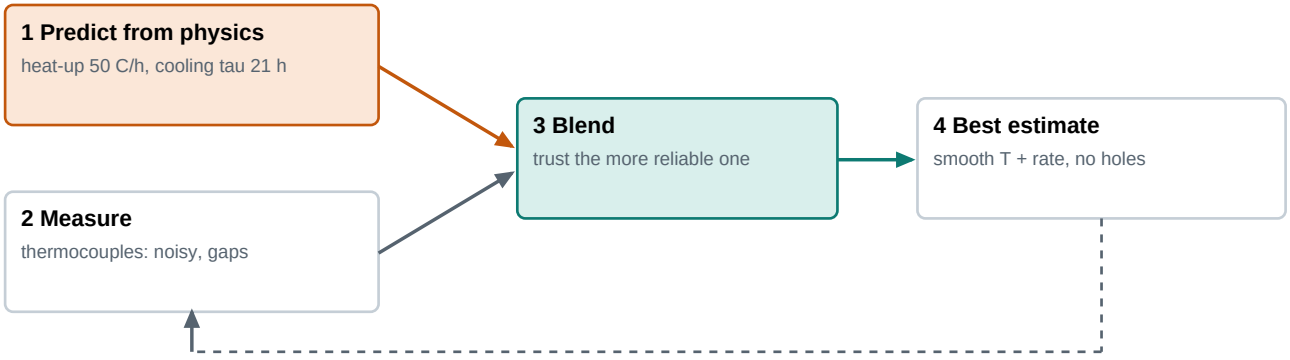
Temperature + heating rate + flows + current → the 50-sensor stage model (86 % right on unseen runs).

Level 3 · Decisions
Let separate judges vote.

Stage rules say "Hot standby", model says "Hot standby" → confident. If they differ → flag for review.

The "predict, then correct" loop

The best-known method is the **Kalman filter**: a running blend of what physics says should happen and what the sensor says happened. Like a navigator before GPS, who guessed the position from speed and heading, then corrected it at each lighthouse.



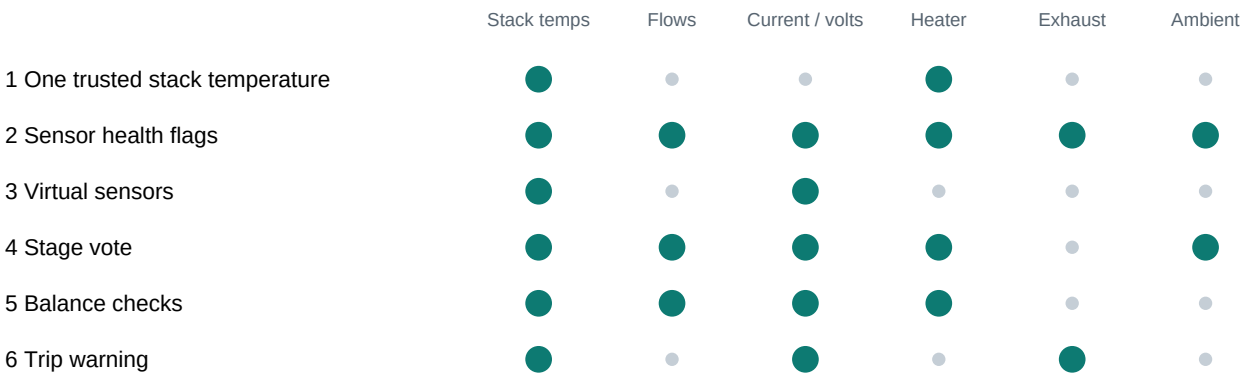
What we already do, without the name

Already in the project	Level	What fusion would add
Ambient proxy = middle of TIR 1,1,1 / 1,2,1 / 1,3 / 1,4,1	1 raw	Add room sensor pi_QIR_10_23, weight by steadiness
Stack spread without TIR 4,6,25	1 raw	Drop bad sensors automatically
Stage rules (operating_states.py)	2 features	Cleaned heating rate for the 5 °C/h Hot standby limit
50-sensor stage model (86 %)	2 features	Vote it against the rules
Rebuilding 106 sensors from 50	2 features	Physics limits so values stay possible

IF YOU REMEMBER ONE THING
Combine same-kind readings for accuracy, different-kind signals for insight, and separate judges for safety.

Where fusion helps this plant

Use	Problem found in the data	Combines	Gives
1 One trusted stack temperature	A stuck thermocouple (TIR 4,6,19 reads 1300 °C), one outside the stack (TIR 4,6,25) and logger dropout minutes distort the stack temperature and its heating rate.	34 TIR 4,6,x + heat-up and cool-down physics	Cleaner Hot standby / Thermal adjustment borders; dropout minutes filled instead of blanked.
2 Sensor health flags	Bad sensors were found by hand: TIR 4,6,19 stuck, 40 package voltages dead since 15 Nov 2025, FIRC 1,1,3 renamed on 17 Mar 2026.	54 twin sensors, 56 groups that move together	A sensor that leaves its group is flagged automatically, every day.
3 Virtual sensors	LIR 7,01 was never logged; 40 package voltages are dead; the 106 other sensors are partly missing from the fake-data cache.	Stack totals EIR 4,s,12/15/18, current, temperatures	Estimated values where no measurement exists, marked as estimates.
4 Stage vote	Rules and the 50-sensor model each make mistakes (model: Thermal adjustment only 0.50 score).	operating_states.py rules + 50-sensor model	Minutes where both agree are trusted; disagreements become a review list and a quality gate for fake data.
5 Balance checks	Fake data copies sensors in pieces, so a block can show current without matching fuel, or heating without heater power.	Fuel flows FIRC 1,2 / 1,3 / 1,5, current EIRC 2,01, heater power	Physics test: electricity drawn needs a fixed amount of hydrogen (Faraday's law), heat in must match temperature rise.
6 Trip warning	8 of 10 part-load runs ended in a safety trip, mostly exhaust TIRZ 8,1,10 at 200 °C; the lowest cell voltage limit (0.72 V) caps power.	Exhaust temperature trend, current, lowest cell voltage	Minutes of warning before a trip; also realistic trip events in fake data.



Which sensor groups each use draws on. Stack thermocouples feed every use.

Link to the fake-data work

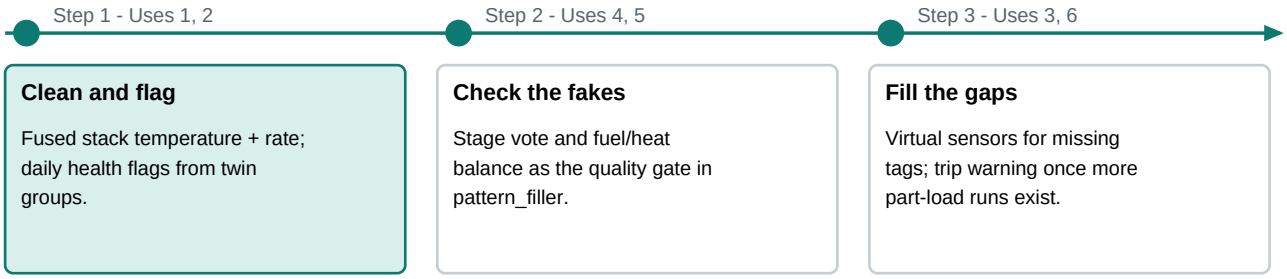
Before filling: uses 1–3 clean the real pieces that get copied. **After filling:** uses 4 and 5 are the quality gate; a fake block below 80 % stage agreement or failing the fuel balance is redrawn.

IF YOU REMEMBER ONE THING

Fusion cleans the real data going into the generator and checks the fake data coming out.

What to build first

Start with the cheap, sure wins every later step depends on; leave the data-hungry uses for when more part-load runs exist.



Limits to respect

Limit	Why it matters	What to do
Fusion cannot create information never measured	Dead package voltages can be estimated from stack totals, not per package with known accuracy	Label estimates as estimates
Twins are not independent witnesses	Side-by-side thermocouples fail together	Count a twin group as one vote
Little data in some stages	Part load 4.5 h in 10 runs; Idle one day	Test on held-out runs F1–F5; trip warning later
Name conflicts	FIRC 1,2: natural gas (stage rules) vs reforming air (ratings)	Settle before the fuel balance

Decisions for you

1. Start with step 1 (fused stack temperature + health flags in data_analysis/stage_signals.py)? Proposed yes.
2. Dropout minutes: fill from the fused estimate (proposed) or keep blanking?
3. Quality gate in pattern_filler.py: redraw failing fake blocks (proposed) or only flag?
4. FIRC 1,2: natural gas or reforming air?

IF YOU REMEMBER ONE THING

Build the trusted stack temperature and health flags first; everything else stands on them.